

3. (a) (i) Friction f between the tyres and the road.

1A

$$f = \frac{mv^2}{r}$$

1M

$$8000 = \frac{1200 v^2}{45 \text{ m}}$$

$$v = 17.3 \text{ m s}^{-1}$$

1A 3

- (ii) Smaller

1A

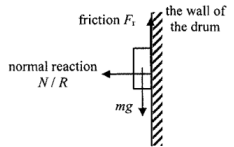
For the same f , $v^2 \propto r$; when r decreases v decreases.

1A 2

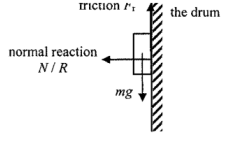
- (b) (Max) friction / coefficient of friction reduced,
not enough to provide the centripetal force / acceleration required for circular motion.
Or Tracking or allowed speed lowered.

1A

1A 2

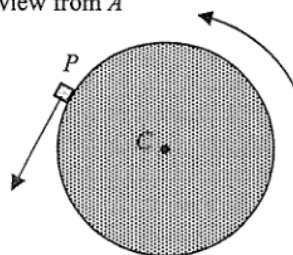
4. (a) (i)		2A	
			2
(ii) Normal reaction (force acting on drum/wall of the drum by the object)		1A	
			1
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ Or $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M	
		1A	
		1M	
		1A	
			2
(ii) centripetal force	$= m\omega^2 r$ $= (2 \times 10^{-5})(78.5)^2(0.25)$ $= 0.03081125 \text{ N} \approx 0.0308 \text{ N}$	1M	
		1A	
			2
(iii) $v = r\omega$	$= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M	
		1A	
			2
(iv) 1		1A	
			1

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		2A	
			2
(ii) Normal reaction (force acting on drum/wall of the drum by the object)		1A	
			1
(b) (i)	$\omega = 78.5 \text{ rad s}^{-1}$ rotation speed = $\frac{78.5}{2\pi} \times 60$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$ Or $T = \frac{2\pi}{\omega} = 0.0800441 \text{ s}$ and $f = \frac{1}{T} = 12.493663 \text{ rev s}^{-1}$ $= 749.619782 \approx 750 \text{ (rev min}^{-1}\text{)}$	1M	
		1A	
		1M	
		1A	
			2
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		1A	
			2
(iii) $v = r\omega$	$= 0.25 \times 78.5$ $= 19.625 \approx 19.6 \text{ m s}^{-1}$	1M	
		1A	
			2
(iv) 1		1A	
			1

4. (a) (i) $F = m\omega^2 r = 0.020 \times 6^2 \times 0.50$
 $= 0.36 \text{ N}$

(ii) Top view from A



(b) (i) same angular speed

(ii) different / smaller magnitude of centripetal acceleration
 $(a_Q < a_P)$
 as Q takes a different / smaller radius ($r_Q < r_P$)

1M
1A
2
1A
1
1A
1
1A
1A
2